

TEMPEST: Soil Gas Well

September 2025 Samples

2026-09-24

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##Run Information

```
cat("Run Information: Brenden Blakley ") #lets you know what section you're in
```

Run Information: Brenden Blakley

```
#set the run date & user name
```

```
sample_year <- "Y2025"
```

```
sample_month <- "September"
```

```
user <- "Brenden Blakley"
```

```
#identify the files you want to read in
```

```
#read in as a list to accommodate multiple runs in a month (.txt file(s))
```

```
GHG_files <- c("Raw Data/20260827_TMP2025_June_Sept.txt", "Raw Data/20260827_TMP2025_Nov_Sept.txt")
```

```
#Metadata file path
```

```
Metadata_path = "Metadata/TEMPEST_SGAS_202509_ALL_WORKING.csv"
```

```
# Define the file path for QAQC log file - NO Need to change just check year
```

```
log_path <- "GC_QAQC_Log.csv"
```

```
check_log_path <- "GC_Check_QAQC_Log.csv"
```

```
# Define final path for data be sure to change year and month and run number
```

```
final_path <- "Processed Data/TMP2025_September.csv"
```

```
##Standard Type ("Actual" or "Half")
```

```
#LTREB, SMARTX, and GENX PW Samples use "Half", others should use "Actual"
```

```
Standard_Type <- "Actual"
```

```
#Did Some Samples require manual area retrieval?
```

```
##List Samples and their areas for each analyte ("CH4", "CO2", or "N2O")
```

```
Manual_Areas <- data.frame(
```

```
  Sample = c("NA"),
```

```
  Area = c("NA"),
```

```
  Analyte = c("NA")
```

```
)
```

```
#Gas in Water Calculations
```

```
#Are All Samples Gas in Water? ("YES" or "NO")
```

```
GIW <- "NO"
```

```
#Are some samples Gas in Water?
```

```
##List Sample IDs
```

```
GIW_Samples <- c()
```

```
# "TempC" column value in Celsius (equilibration temperature) and volume of gas (in liters) in syringe
```

```
TempC = 25 #Equilibration Temperature (Celsius)
```

```
volume_g = 0.015 #volume of gas (in liters) in syringe when equilibrated
```

```
volume_w = 0.015 #volume of water (in liters) in syringe when equilibrated (should be equal to volume_g)
```

```
#record any notes about the run or anything other info here:
```

```
run_notes <- "Run 9/02 - Blank1,2,4,5,11 -> ChkStd -- Corrected"

#Collection_Dates = "Raw Data/Sample_Collection_Dates_2024.csv"

cat(run_notes)
```

```
## Run 9/02 - Blank1,2,4,5,11 -> ChkStd -- Corrected
```

```
##Set Up
```

```
##Read in metadata and create similar sample IDs for matching to samples
```

```
##Read In Data Files
```

```
#Get Location of Each Sample in Main Data Frame
```

```
#Turn Each Sample into it's own Data Frame
```

```
#Get Only Area Tables and Retrieve Info
```

```
##QAQC Peak Count and Retention Time
```

```
## [1] "If there are too many peaks there is likely a problem with the sample or run and the chromatogr
```

```
## [1] "FID: Oppm; 1" "FID: Std_10K_3mL; 1"
## [3] "FID: Std_10K_6mL; 1" "FID: Std_10K_12mL; 1"
## [5] "FID: Std_10K_24mL; 1" "FID: Blank3; 0"
## [7] "FID: 2025_June_42; 3" "FID: 2025_June_59; 3"
## [9] "FID: Blank6; 1" "FID: Blank7; 1"
## [11] "FID: 2025_September_12; 3" "FID: Blank8; 1"
## [13] "FID: Blank10; 1" "FID: Blank12; 1"
## [15] "FID: Blank1; 1" "FID: Blank2; 1"
## [17] "FID: Oppm; 1" "FID: Std_10K_3mL; 1"
## [19] "FID: Std_10K_6mL; 1" "FID: Std_10K_12mL; 1"
## [21] "FID: Std_10K_24mL; 1" "FID: Blank3; 1"
## [23] "FID: Blank4; 1" "FID: Blank5; 1"
## [25] "FID: Blank6; 1" "FID: Blank7; 1"
## [27] "FID: ChkStd_HighTri_12mL_5; 3" "FID: Blank8; 1"
## [29] "FID: Blank9; 1" "FID: Blank10; 1"
## [31] "FID: Blank11; 1"
```

```
## [1] "ECD: Std_10K_3mL; 0" "ECD: Std_10K_6mL; 0" "ECD: Std_10K_12mL; 0"
## [4] "ECD: Std_10K_24mL; 0" "ECD: Blank3; 0" "ECD: Blank6; 0"
## [7] "ECD: Blank7; 0" "ECD: Blank8; 0" "ECD: Blank10; 0"
## [10] "ECD: Blank12; 0" "ECD: Blank1; 0" "ECD: Blank2; 0"
## [13] "ECD: Oppm; 0" "ECD: Std_10K_3mL; 0" "ECD: Std_10K_6mL; 0"
## [16] "ECD: Std_10K_12mL; 0" "ECD: Std_10K_24mL; 0" "ECD: Blank3; 0"
## [19] "ECD: Blank4; 0" "ECD: Blank5; 0" "ECD: Blank6; 0"
## [22] "ECD: Blank7; 0" "ECD: Blank8; 0" "ECD: Blank9; 0"
## [25] "ECD: Blank10; 0" "ECD: Blank11; 0"
```

```
## [1] "If RTime is wrong check that correct number of peaks were found, if not a peak may have been mi
```

```
## [1] "CH4: "
```

```
## [1] "CO2:  "
```

```
## [1] "N2O:  "
```

```
##Format Data
```

```
##Add Standard Concentrations
```

```
##Finalize Data
```

1 Assess standard curves CH4

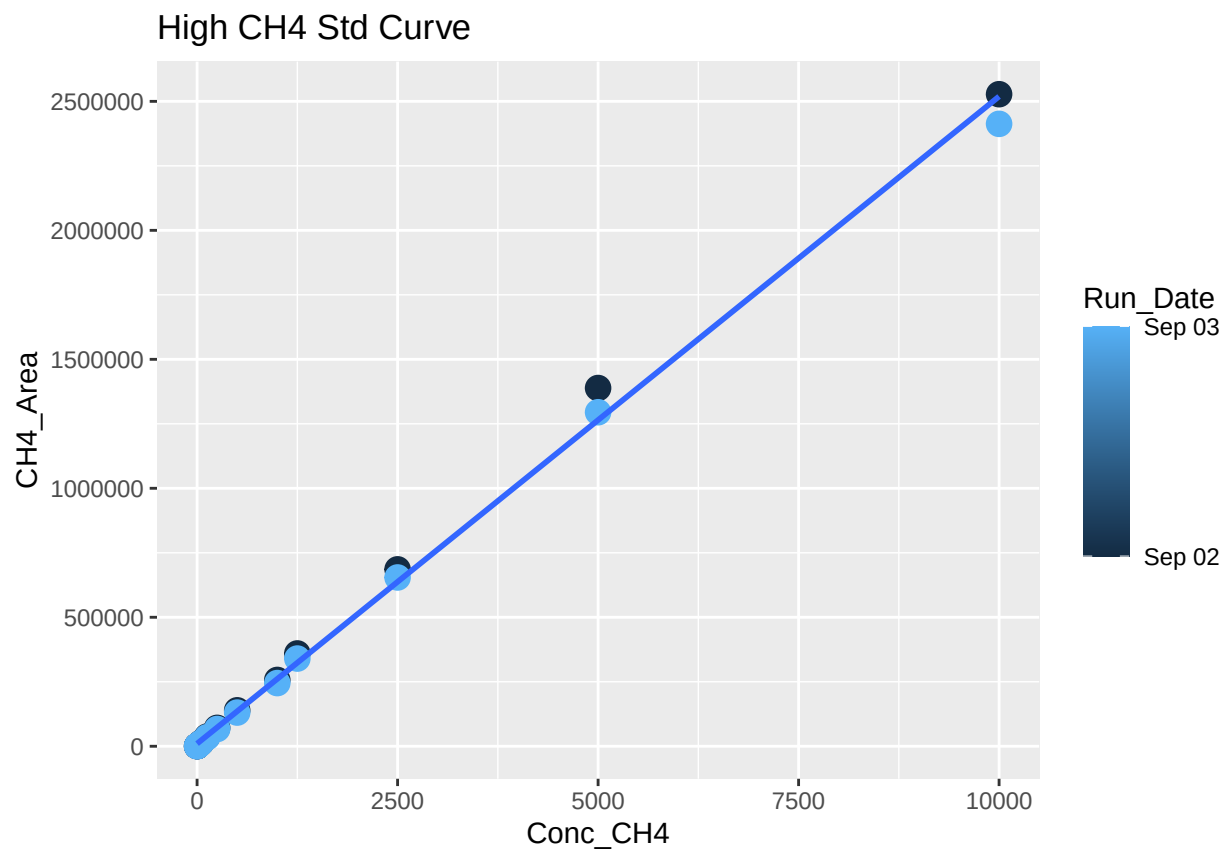
```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: The following aesthetics were dropped during statistical transformation:
```

```
## colour.
```

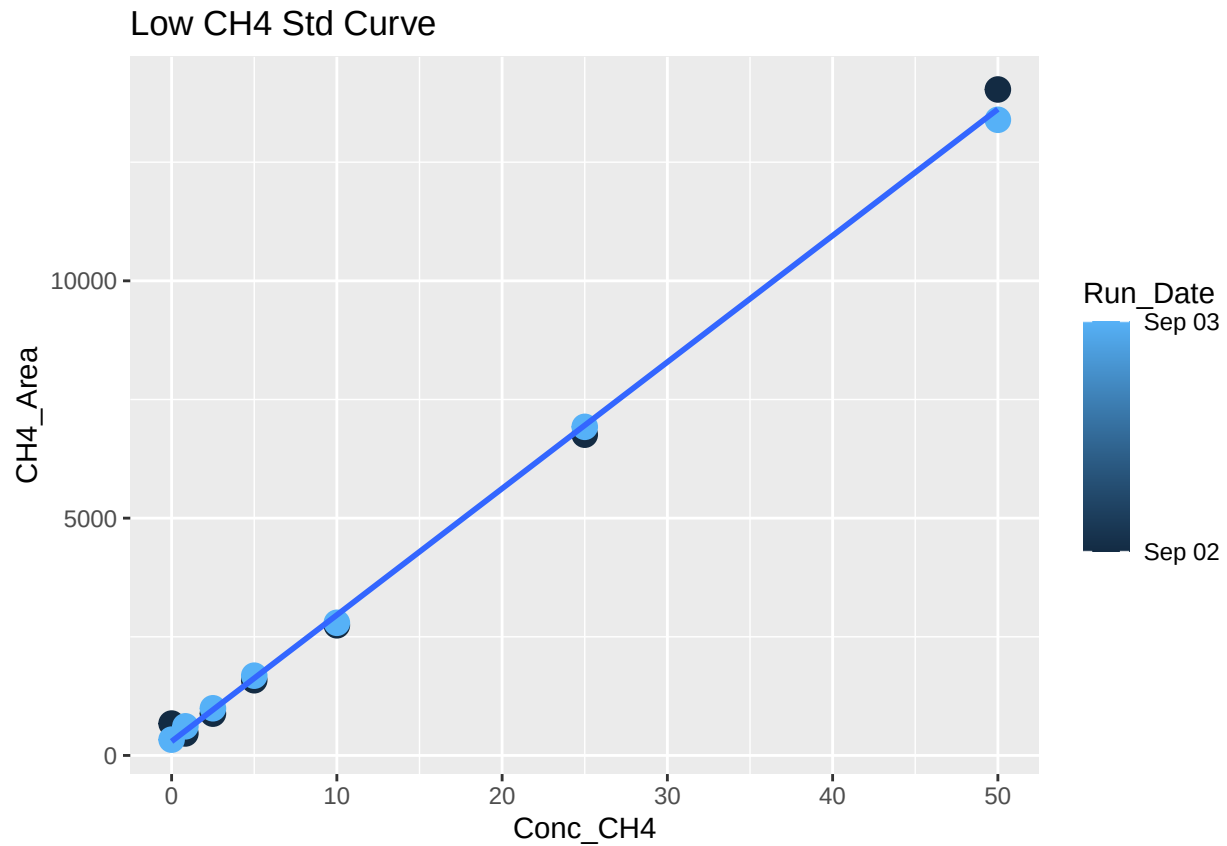
```
## i This can happen when ggplot fails to infer the correct grouping structure in  
## the data.
```

```
## i Did you forget to specify a 'group' aesthetic or to convert a numerical  
## variable into a factor?
```

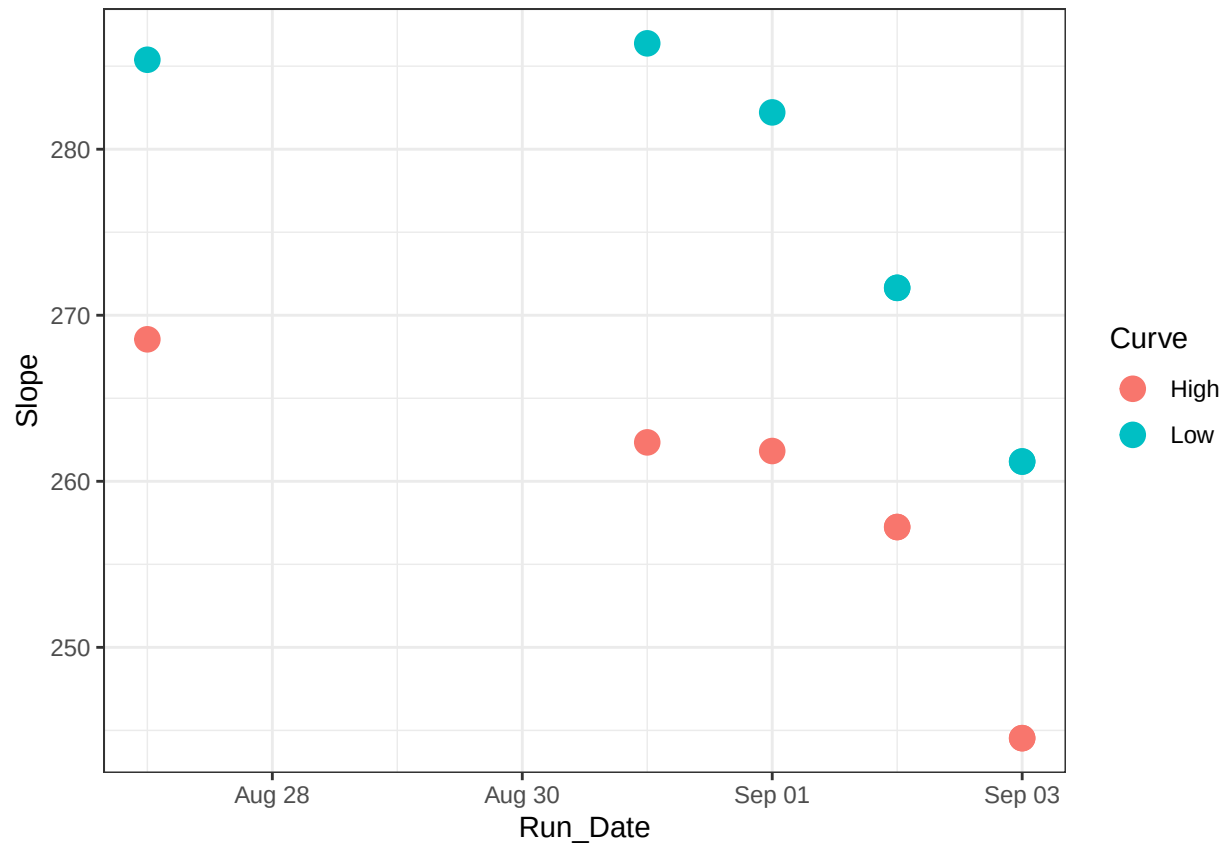


```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: The following aesthetics were dropped during statistical transformation:
## colour.
## i This can happen when ggplot fails to infer the correct grouping structure in
## the data.
## i Did you forget to specify a 'group' aesthetic or to convert a numerical
## variable into a factor?
```



##	Run_Date	Curve	R2	Slope	Intercept	Analyte
## 1	2026-08-27	High	0.9986559	268.551423	9535.1095	CH4
## 2	2026-08-27	Low	0.9995601	285.387517	115.8345	CH4
## 3	2026-08-27	Full	0.9984309	2.396263	254.4449	CO2
## 4	2026-08-27	High	0.9889439	1014.018375	3704.2202	N2O
## 5	2026-08-27	Low	0.9986226	1454.680571	214.6734	N2O
## 6	2026-08-31	High	0.9987205	262.344419	9865.8585	CH4



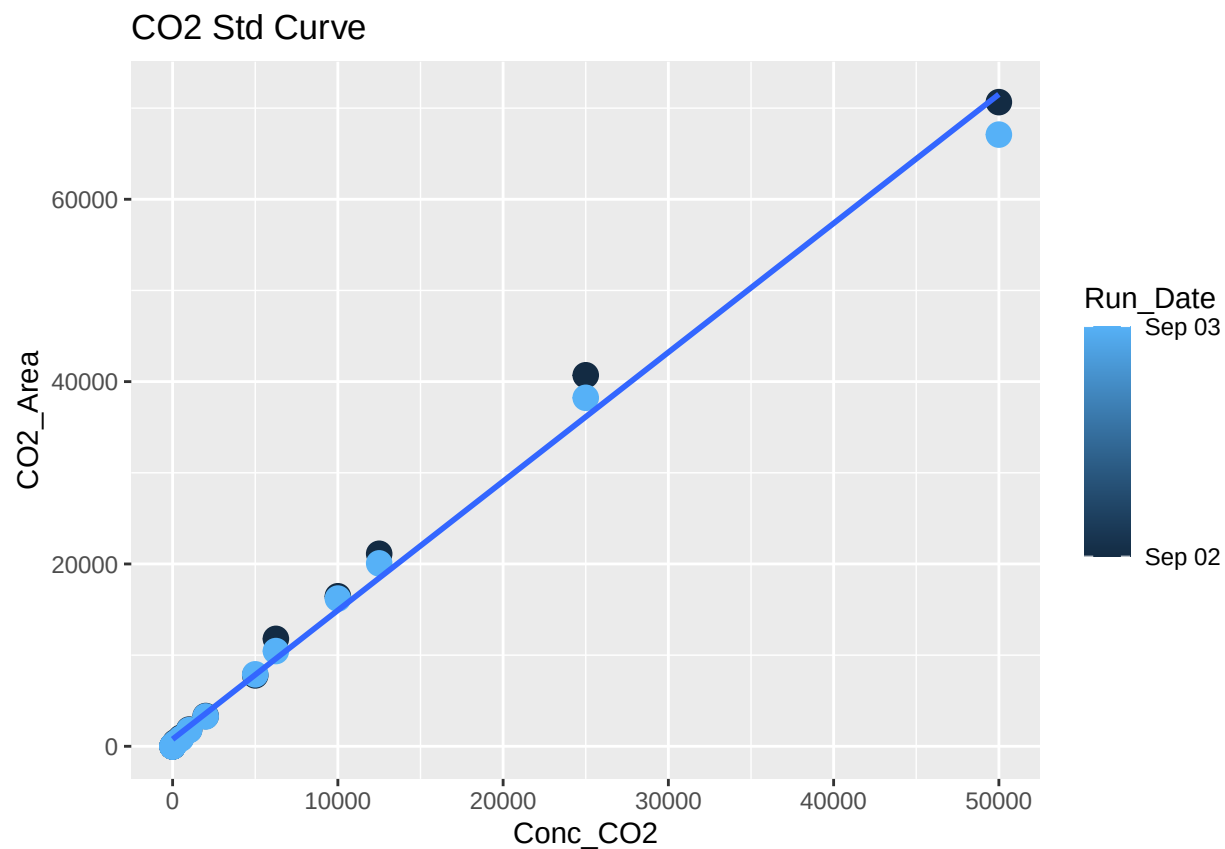
```
## [1] "High_R2 is Good-2026-09-02" "High_R2 is Good-2026-09-03"
```

```
## [1] "Low_R2 is Good-2026-09-02" "Low_R2 is Good-2026-09-03"
```

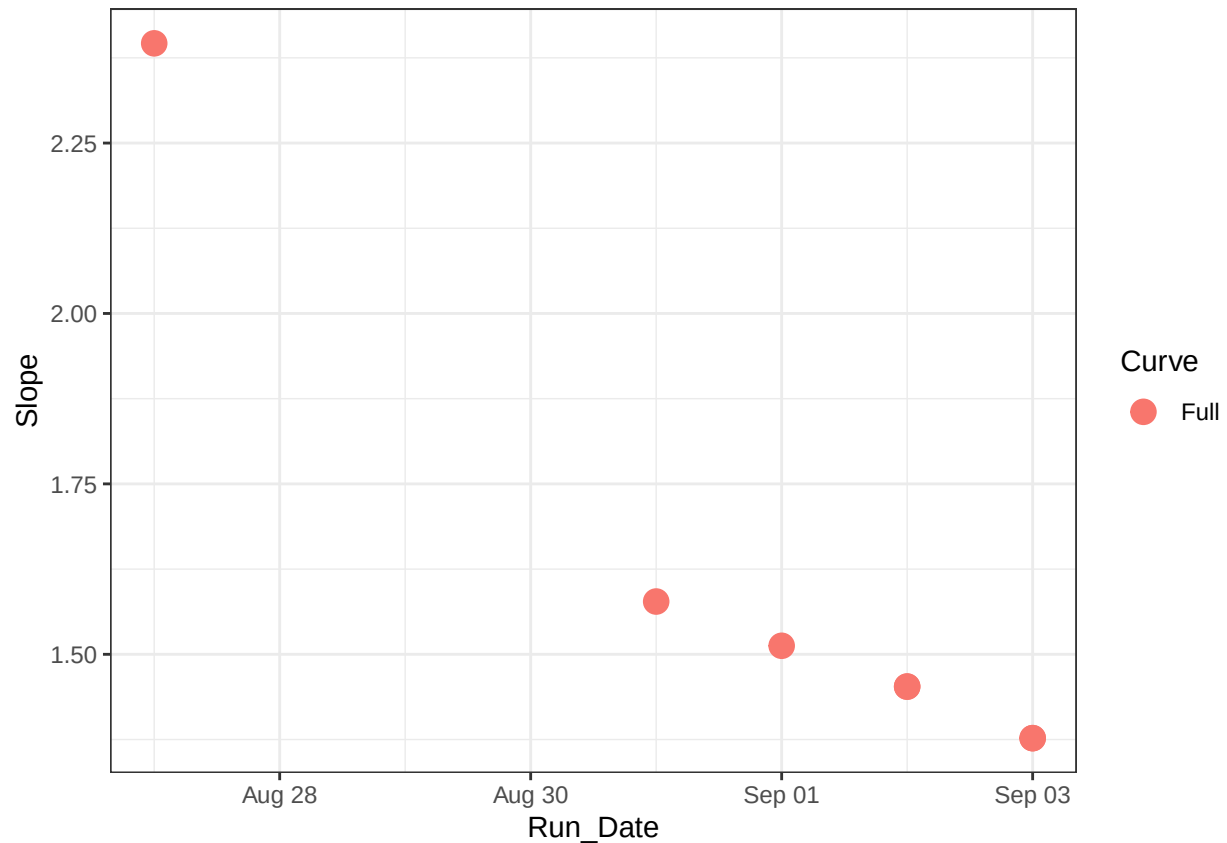
2 Assess standard curves CO2

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: The following aesthetics were dropped during statistical transformation:
## colour.
## i This can happen when ggplot fails to infer the correct grouping structure in
## the data.
## i Did you forget to specify a 'group' aesthetic or to convert a numerical
## variable into a factor?
```



##	Run_Date	Curve	R2	Slope	Intercept	Analyte
## 1	2026-08-27	High	0.9986559	268.551423	9535.1095	CH4
## 2	2026-08-27	Low	0.9995601	285.387517	115.8345	CH4
## 3	2026-08-27	Full	0.9984309	2.396263	254.4449	CO2
## 4	2026-08-27	High	0.9889439	1014.018375	3704.2202	N2O
## 5	2026-08-27	Low	0.9986226	1454.680571	214.6734	N2O
## 6	2026-08-31	High	0.9987205	262.344419	9865.8585	CH4

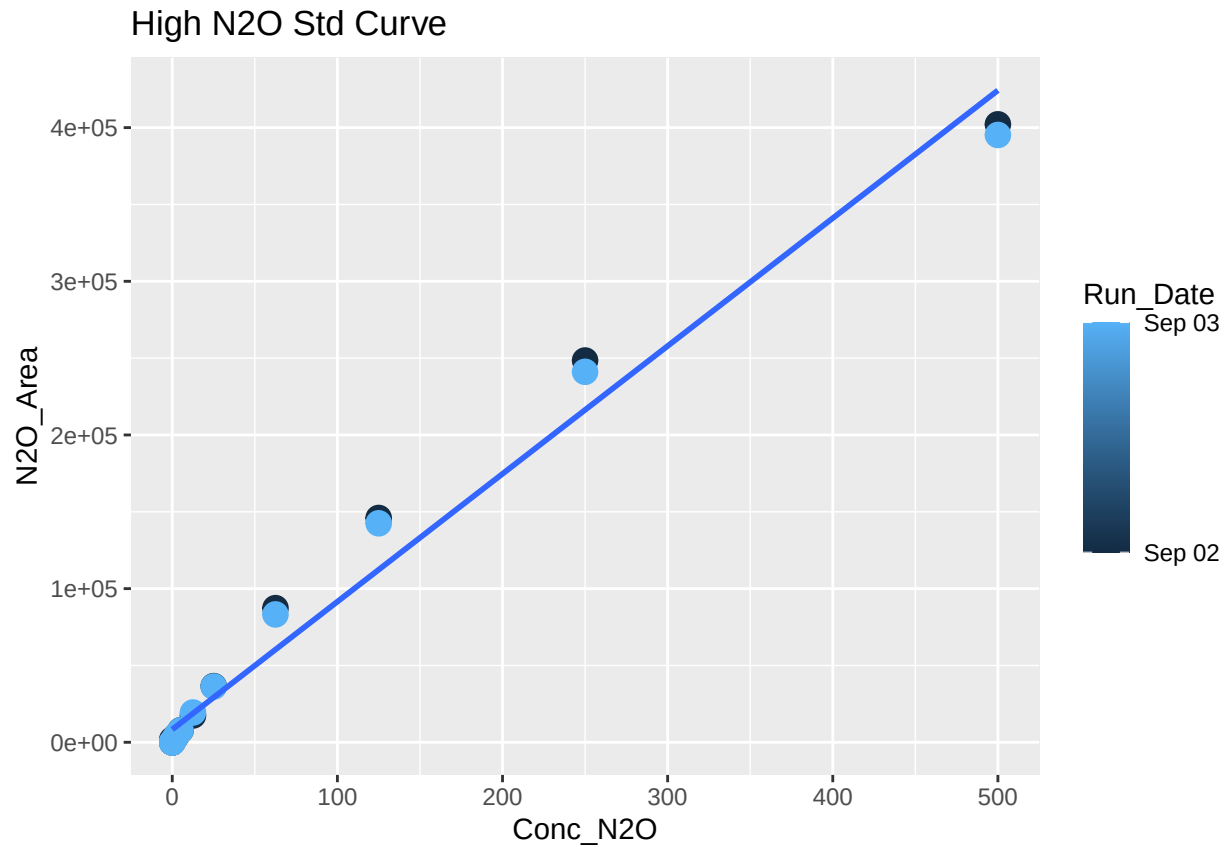


```
## [1] "C02_R2 is Good-2026-09-02" "C02_R2 is Good-2026-09-03"
```

3 Assess standard curves N2O

```
## 'geom_smooth()' using formula = 'y ~ x'
```

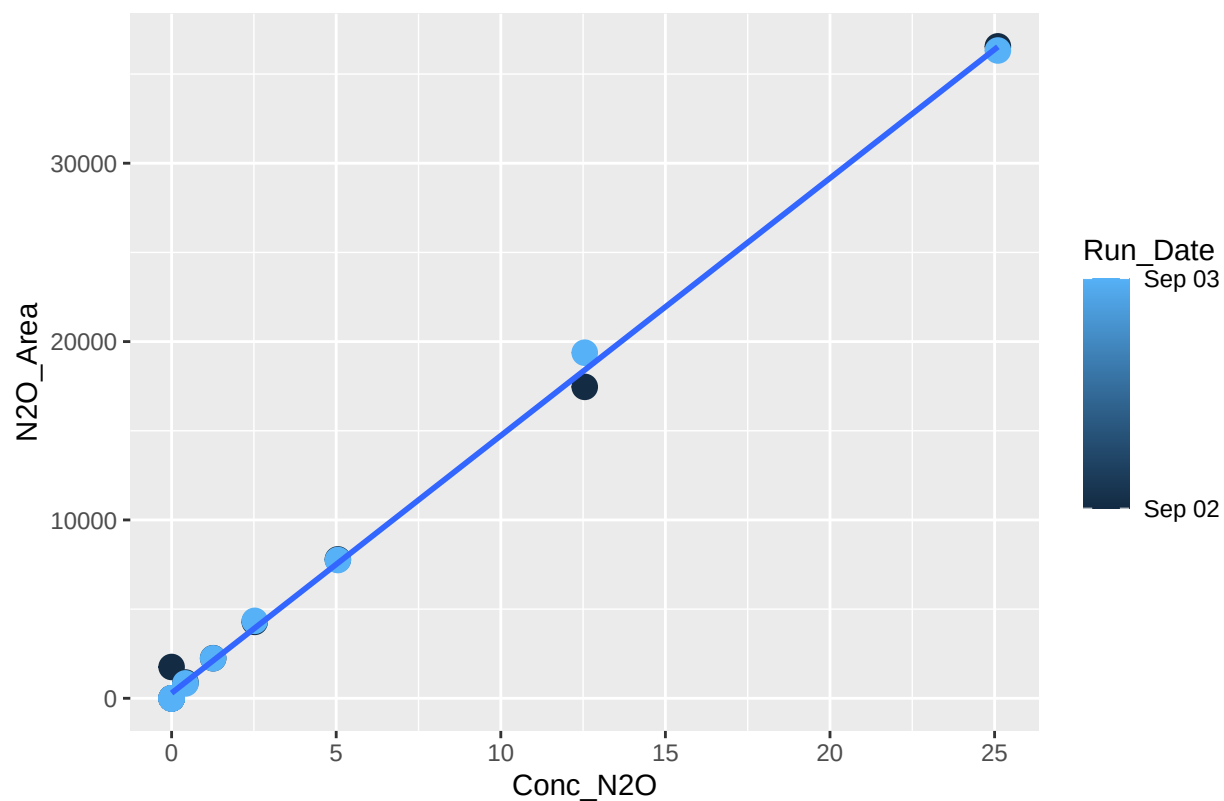
```
## Warning: The following aesthetics were dropped during statistical transformation:
## colour.
## i This can happen when ggplot fails to infer the correct grouping structure in
## the data.
## i Did you forget to specify a 'group' aesthetic or to convert a numerical
## variable into a factor?
```



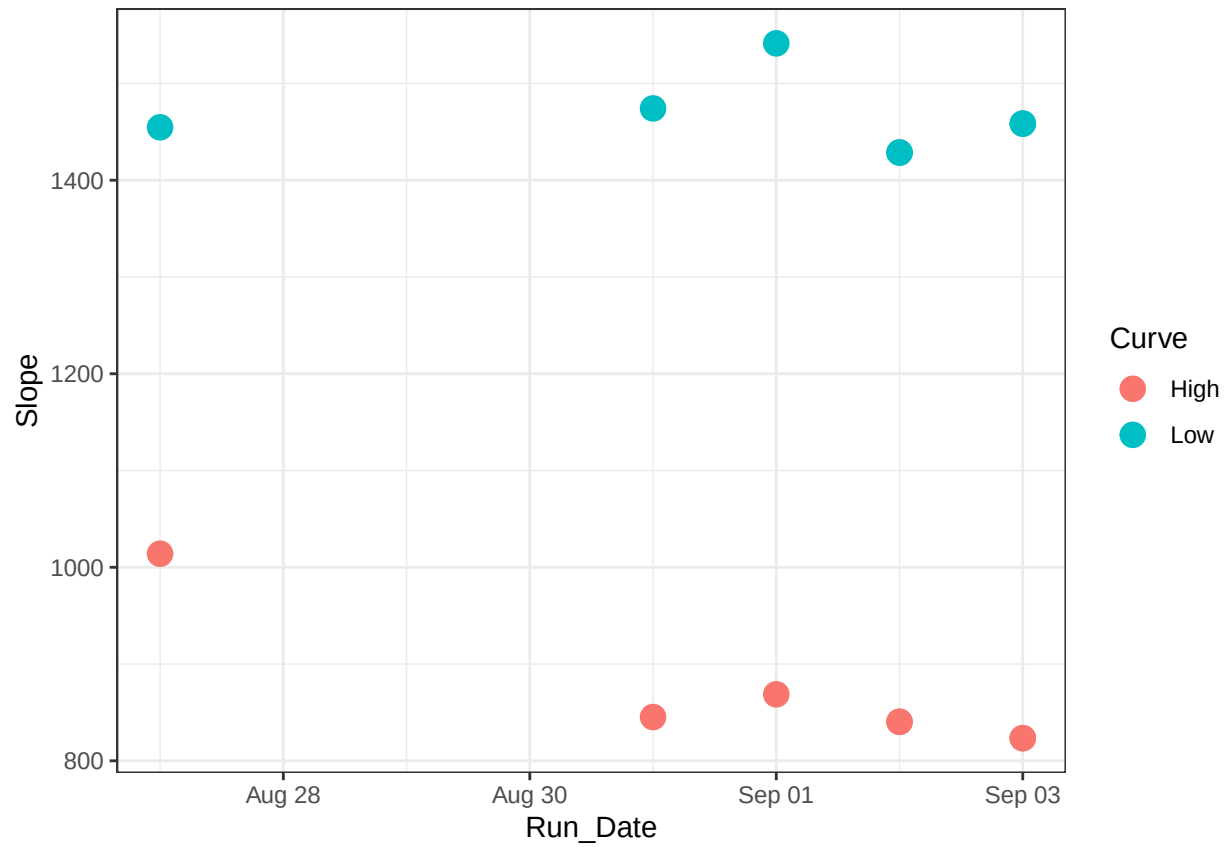
```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: The following aesthetics were dropped during statistical transformation:  
## colour.  
## i This can happen when ggplot fails to infer the correct grouping structure in  
## the data.  
## i Did you forget to specify a 'group' aesthetic or to convert a numerical  
## variable into a factor?
```

Low N2O Std Curve



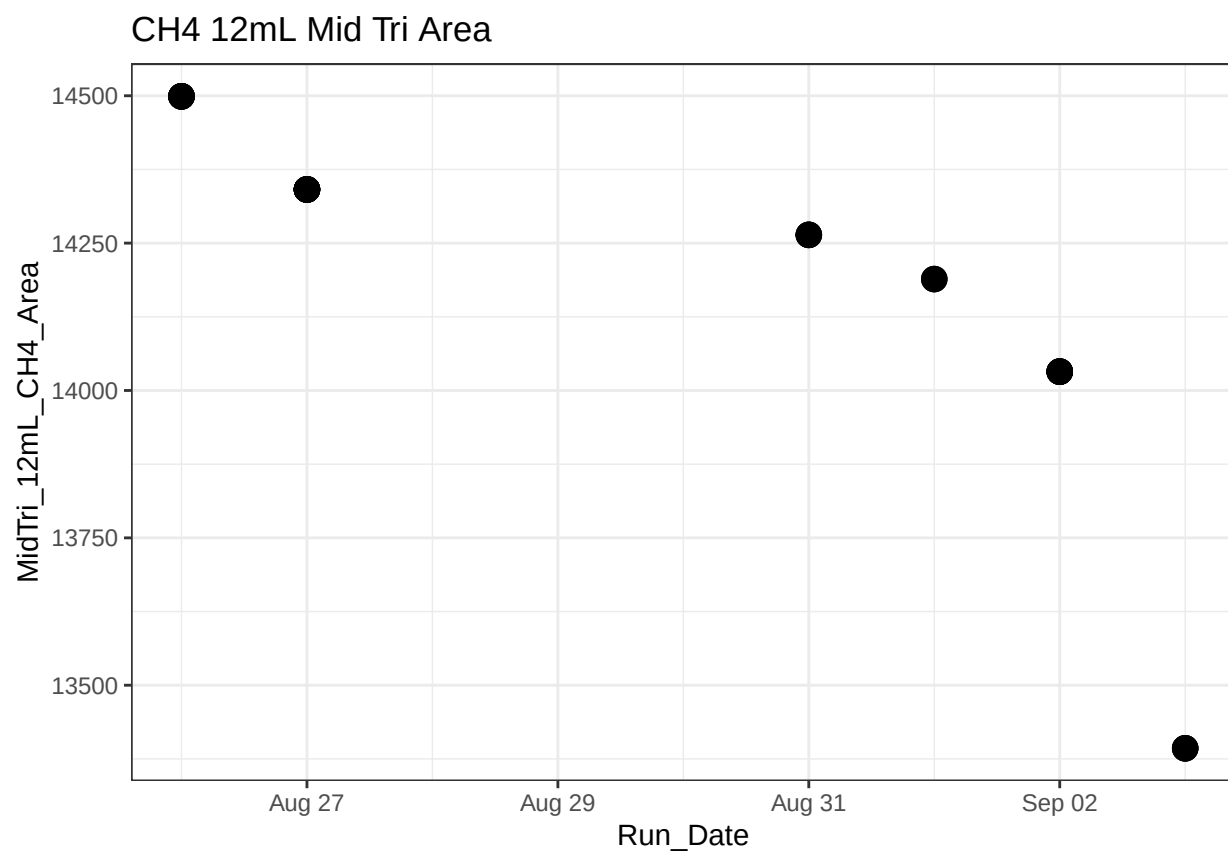
##	Run_Date	Curve	R2	Slope	Intercept	Analyte
## 1	2026-08-27	High	0.9986559	268.551423	9535.1095	CH4
## 2	2026-08-27	Low	0.9995601	285.387517	115.8345	CH4
## 3	2026-08-27	Full	0.9984309	2.396263	254.4449	CO2
## 4	2026-08-27	High	0.9889439	1014.018375	3704.2202	N2O
## 5	2026-08-27	Low	0.9986226	1454.680571	214.6734	N2O
## 6	2026-08-31	High	0.9987205	262.344419	9865.8585	CH4

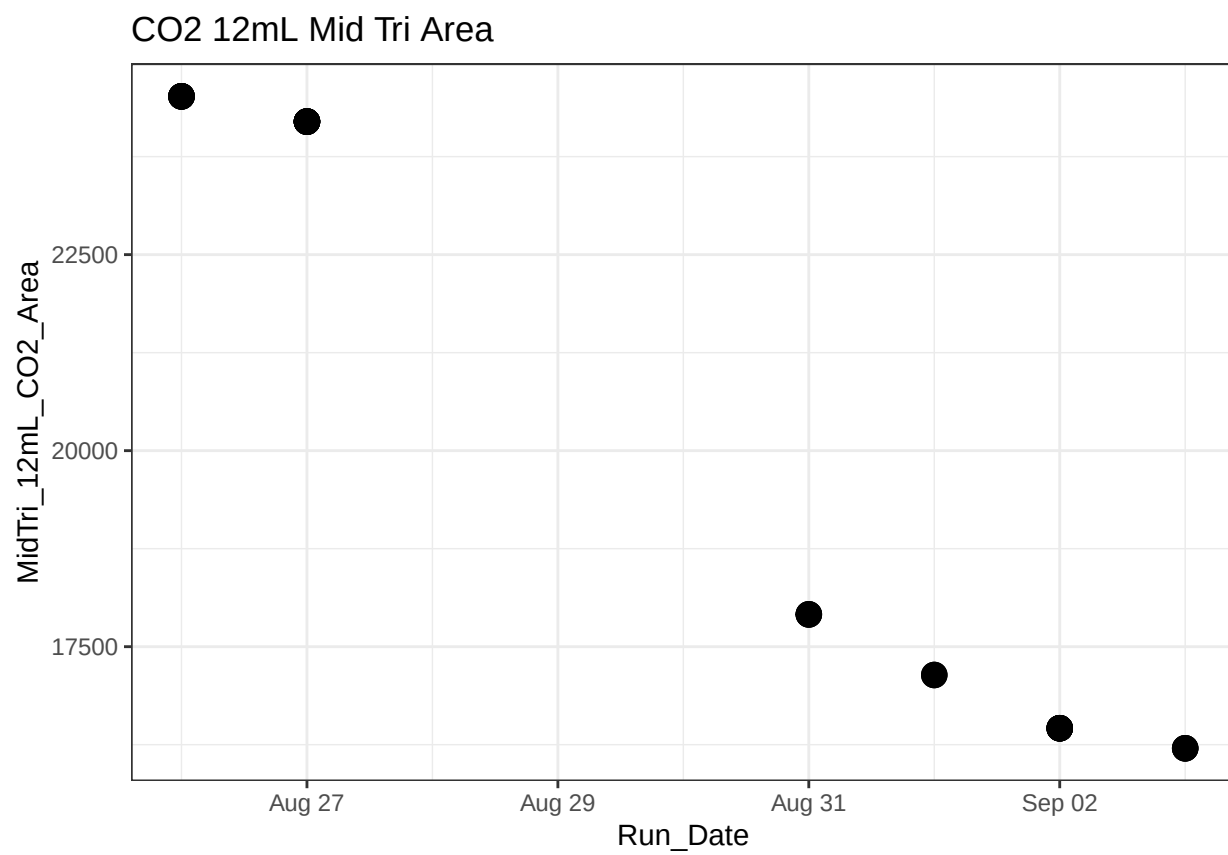


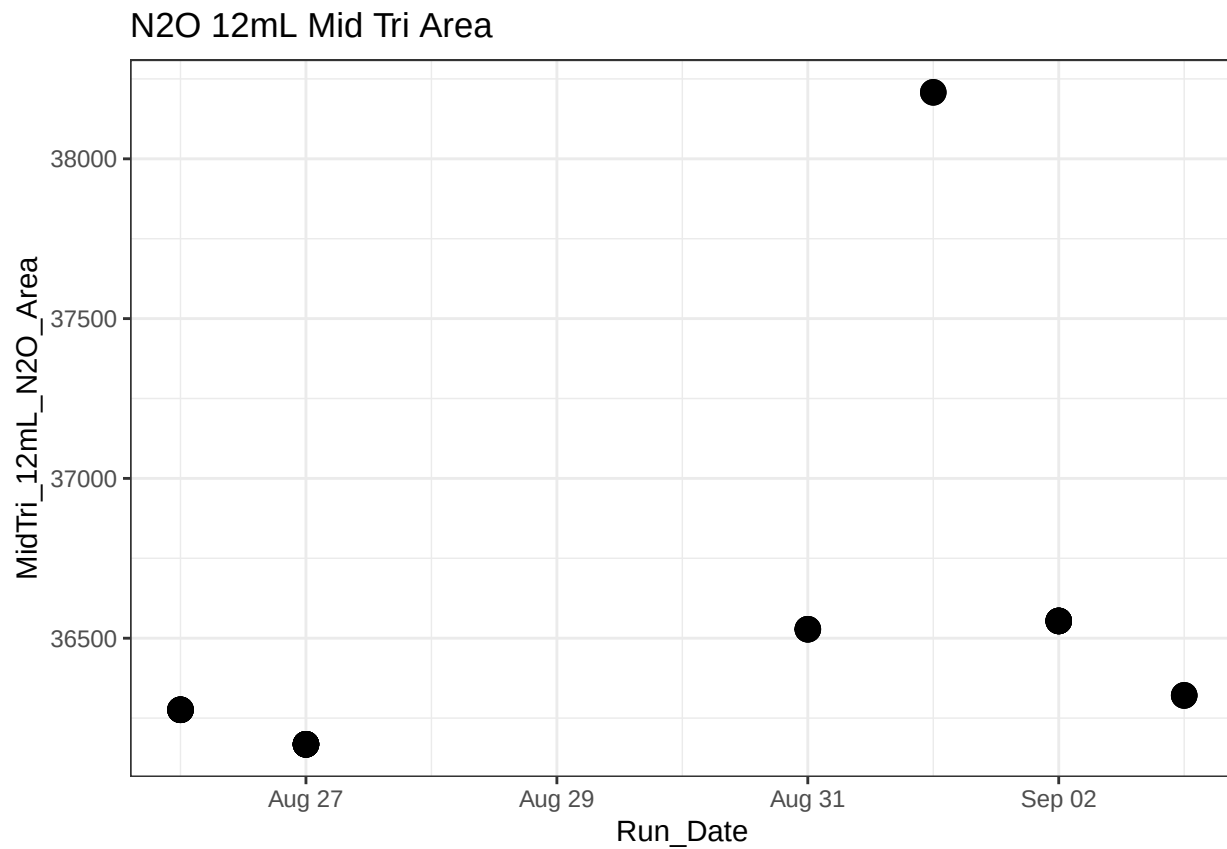
```
## [1] "High_R2 is Bad-2026-09-02" "High_R2 is Bad-2026-09-03"
```

```
## [1] "Low_R2 is Good-2026-09-02" "Low_R2 is Good-2026-09-03"
```

```
#Plot Check Areas over Time
```







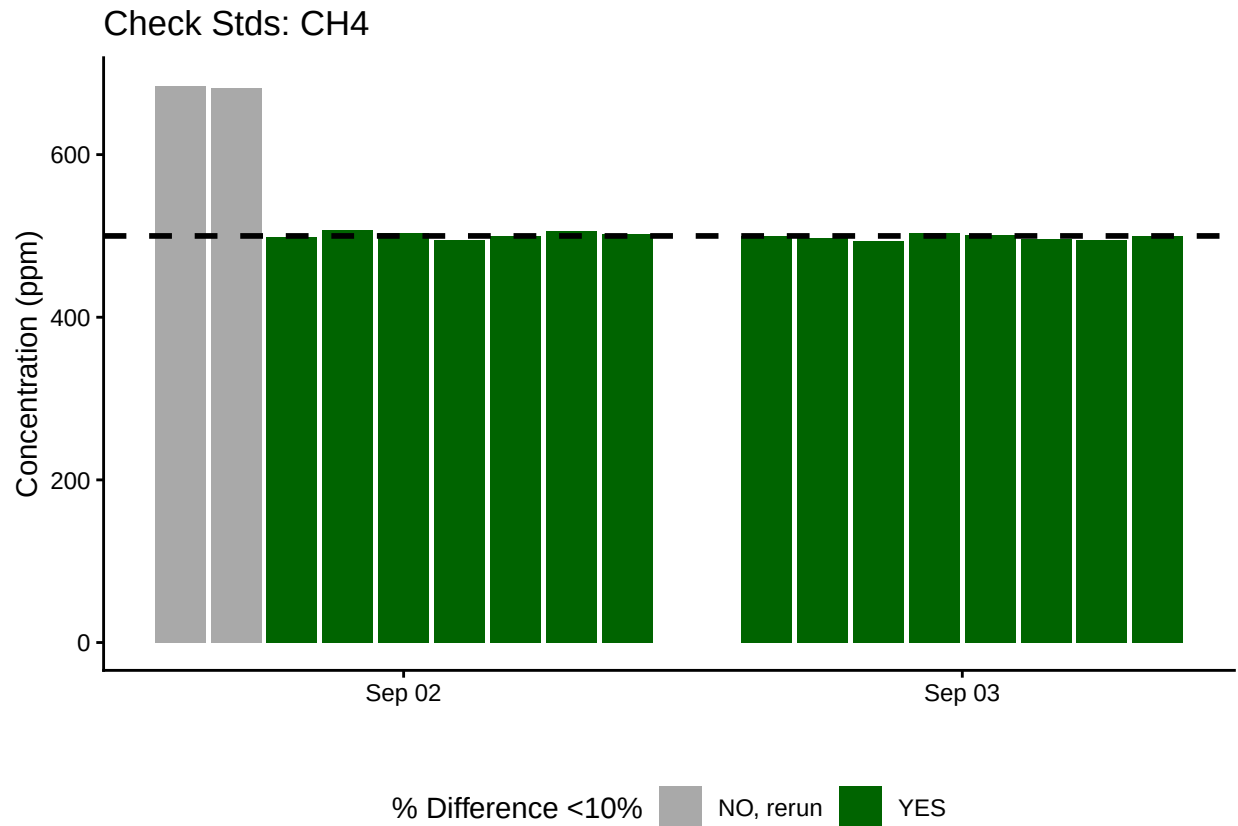
3.1 Now calculate the CH₄ in ppm

3.2 Now calculate the CO₂ in ppm

3.3 Now calculate the N₂O in ppm

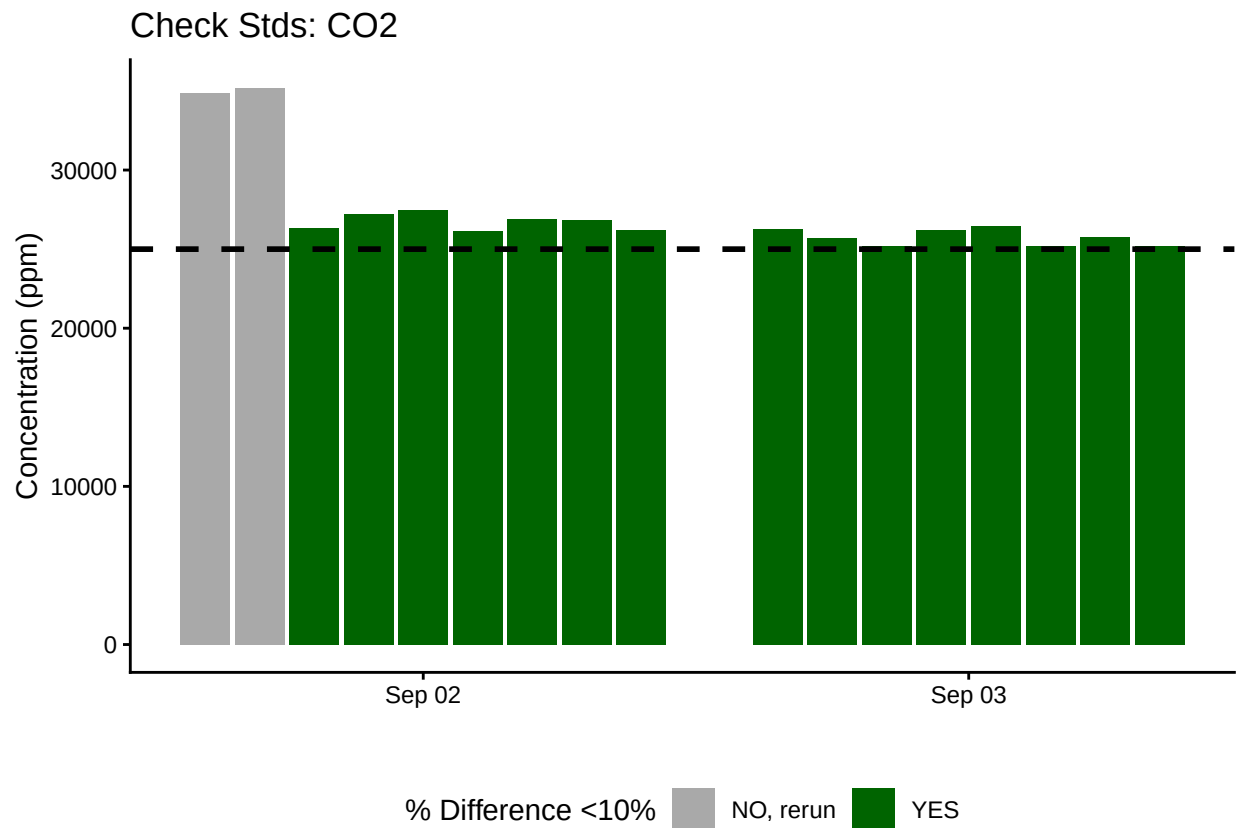
4 Aanlayze the CH₄ Check Standards

>60% of CH₄ Check Standards are within range of expected concentration - PROCEED



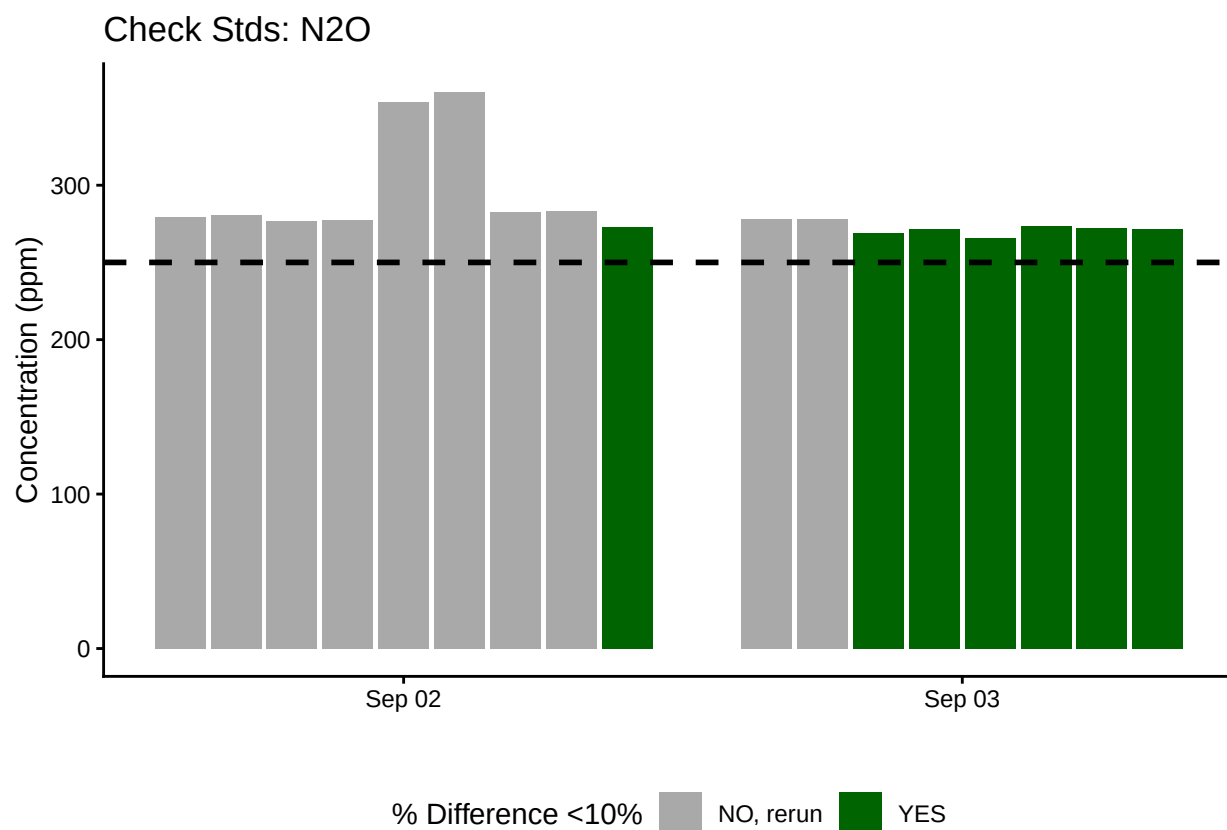
5 Aanlayze the CO2 Check Standards

>60% of CO2 Check Standards are within range of expected concentration - PROCEED



6 Aanlayze the N2O Check Standards

<60% of N2O Check Standards are within range of expected concentration - REASSESS



7 Analyze the CH₄ Blanks

Assess CH₄ Blanks

>60% of CH₄ Blanks are below the lower 25% quartile of samples - PROCEED

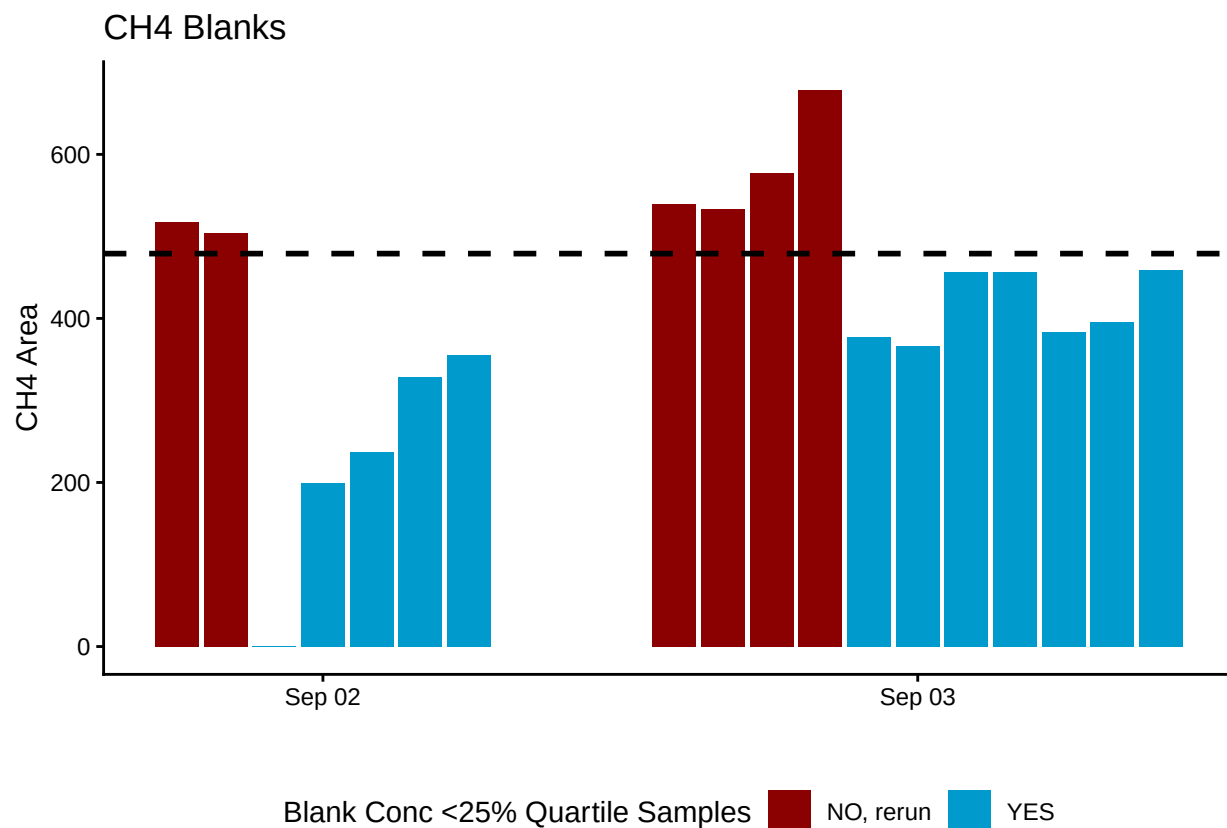


Table 1: Mean Area of Blanks

Test	Blank_Mean_Area
CH4	409.2778

8 Analyze the CO2 Blanks

```
## Assess CO2 Blanks
```

```
## >60% of CO2 Blanks are below the lower 25% quartile of samples - PROCEED
```

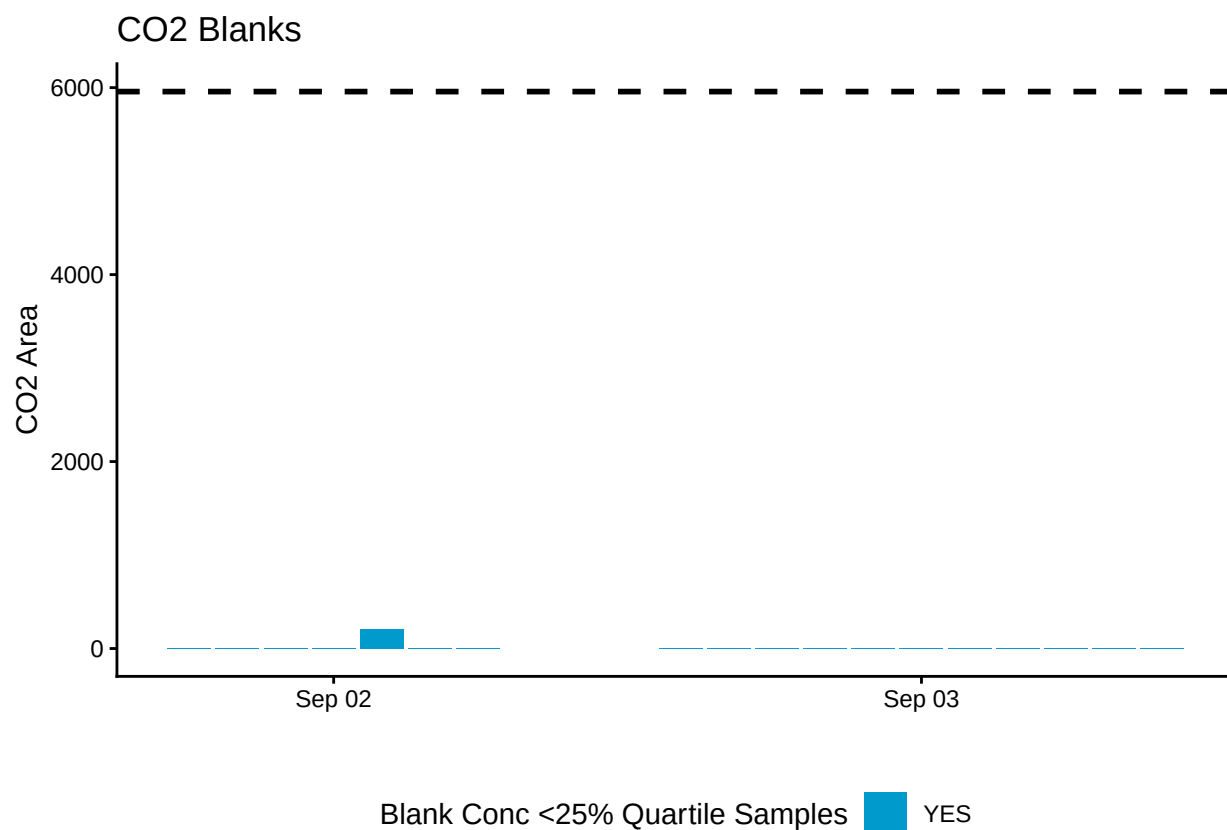


Table 2: Mean Area of Blanks

Test	Blank_Mean_Area
CO2	11.8333

9 Analyze the N2O Blanks

Assess N2O Blanks

>60% of N2O Blanks are below the lower 25% quartile of samples - PROCEED

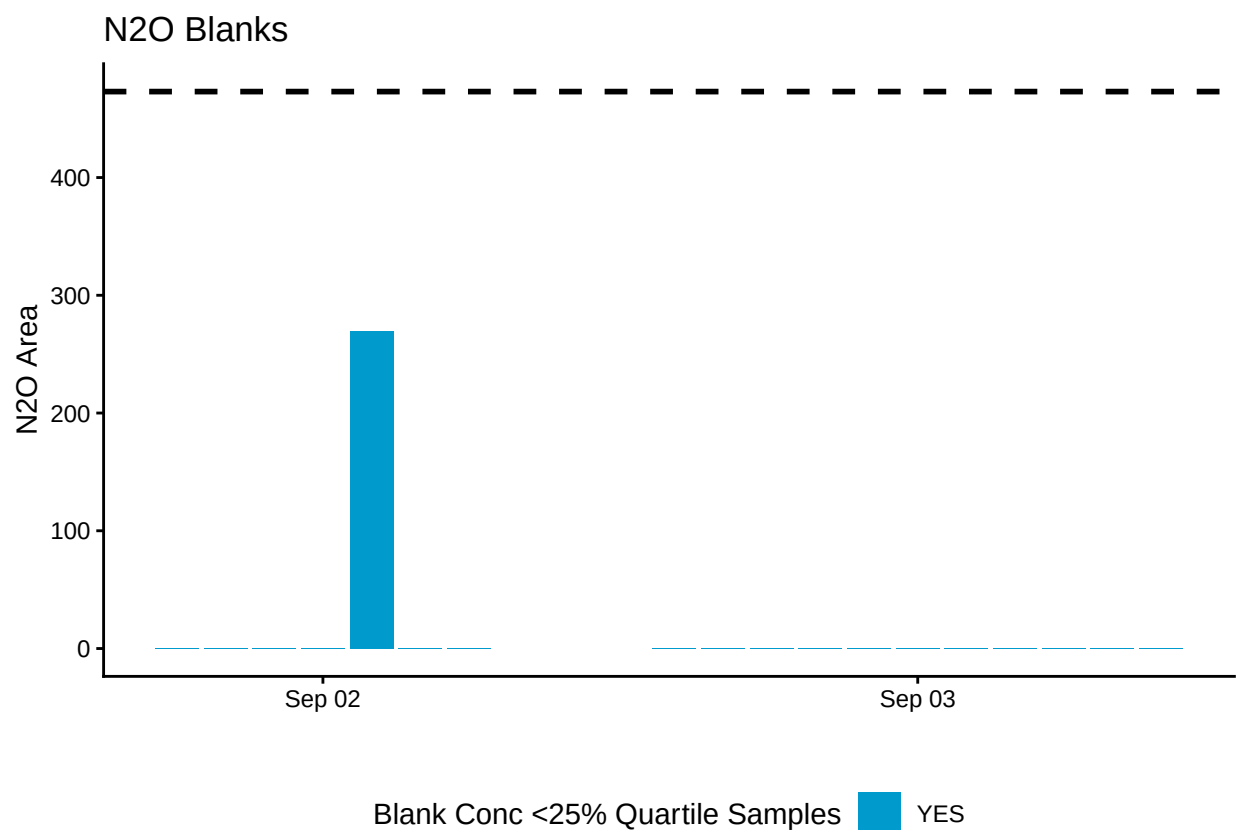
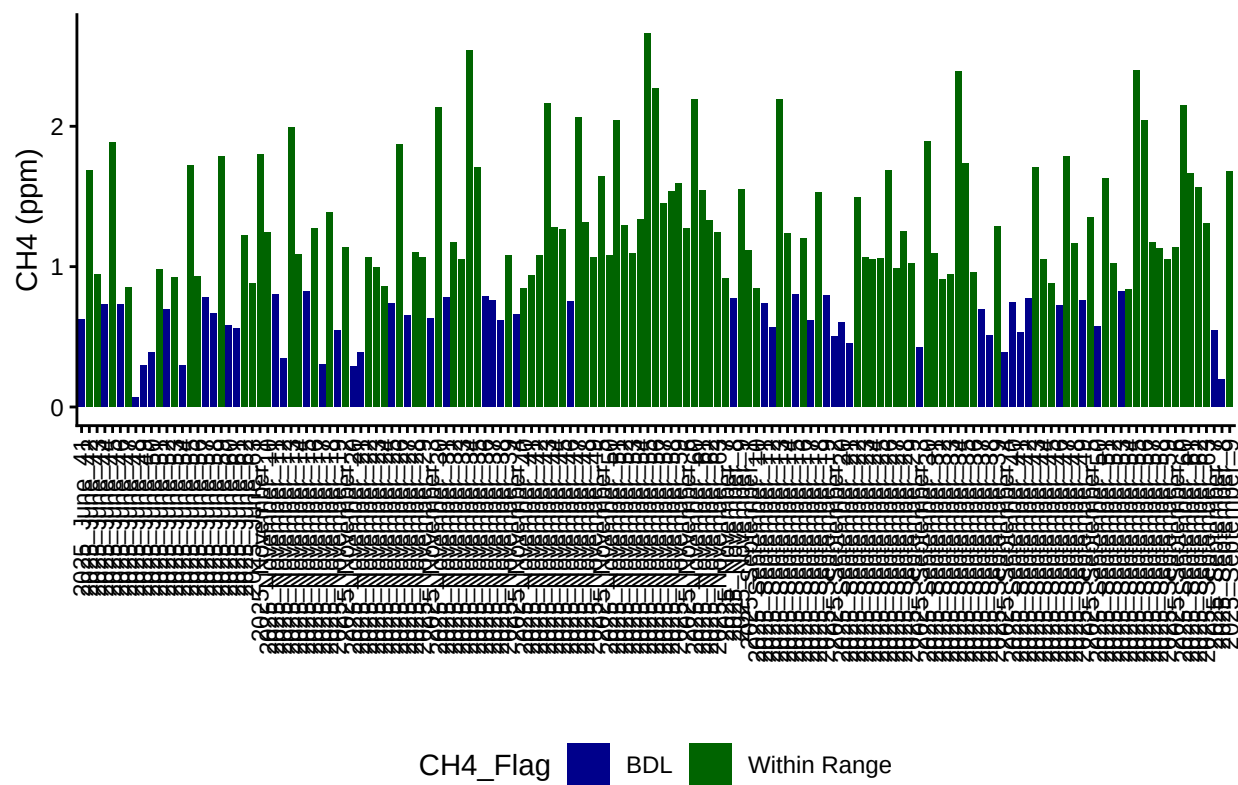


Table 3: Mean Area of Blanks

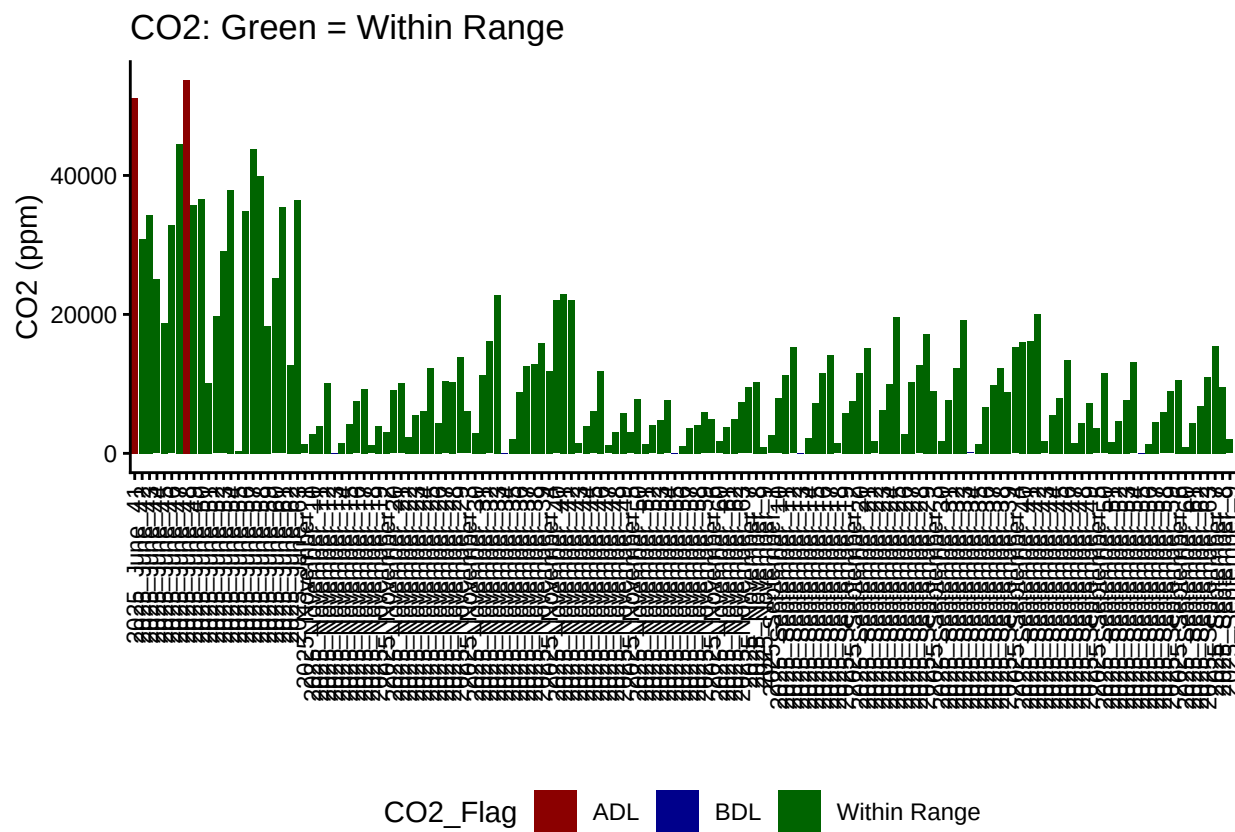
Test	Blank_Mean_Area
N2O	15

10 Check Range - CH4

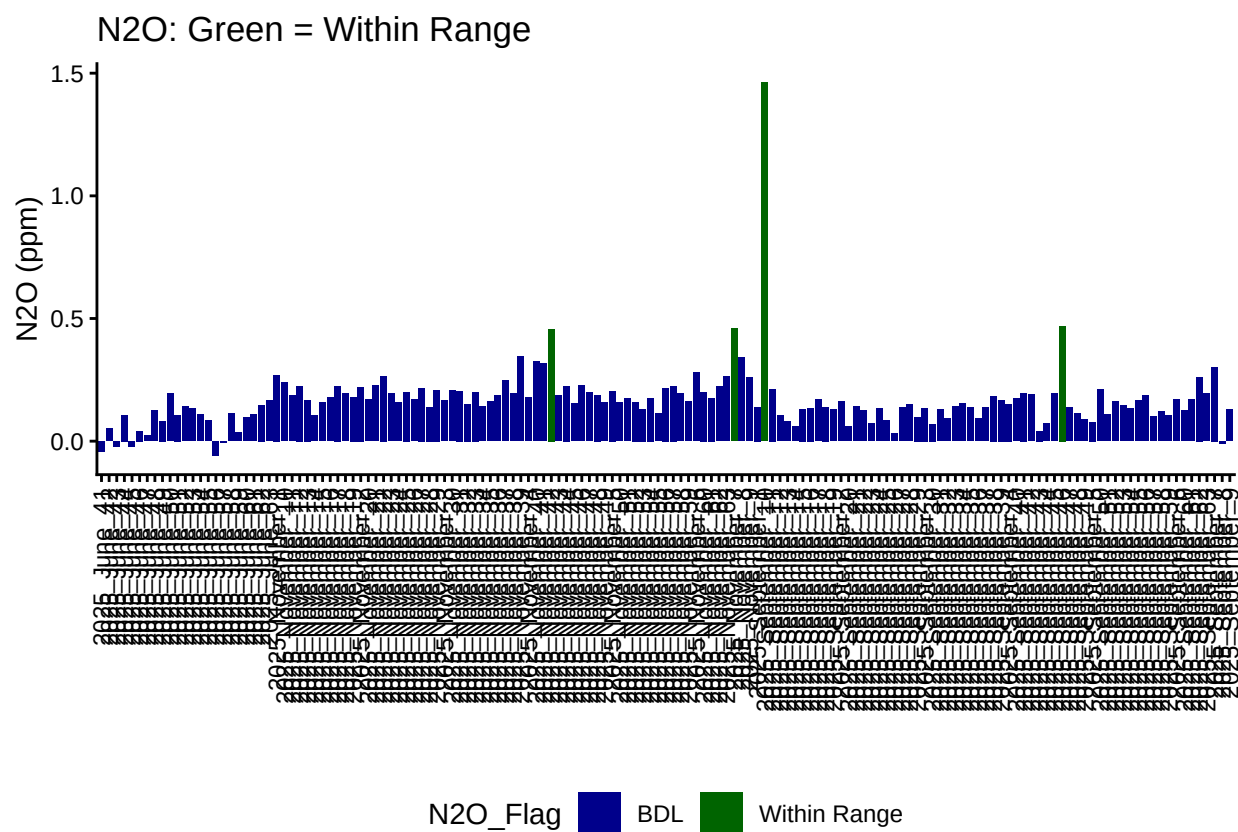
CH4: Green = Within Range



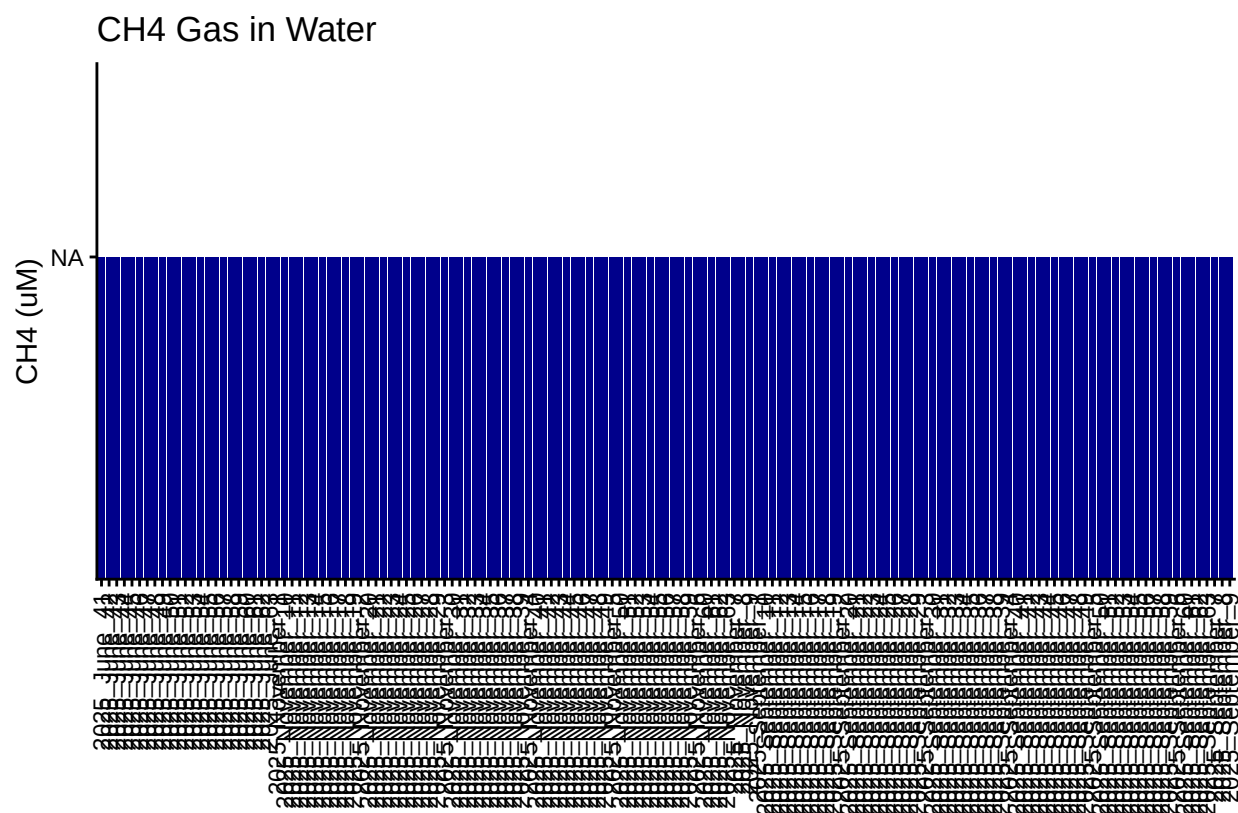
11 Check Range - CO2



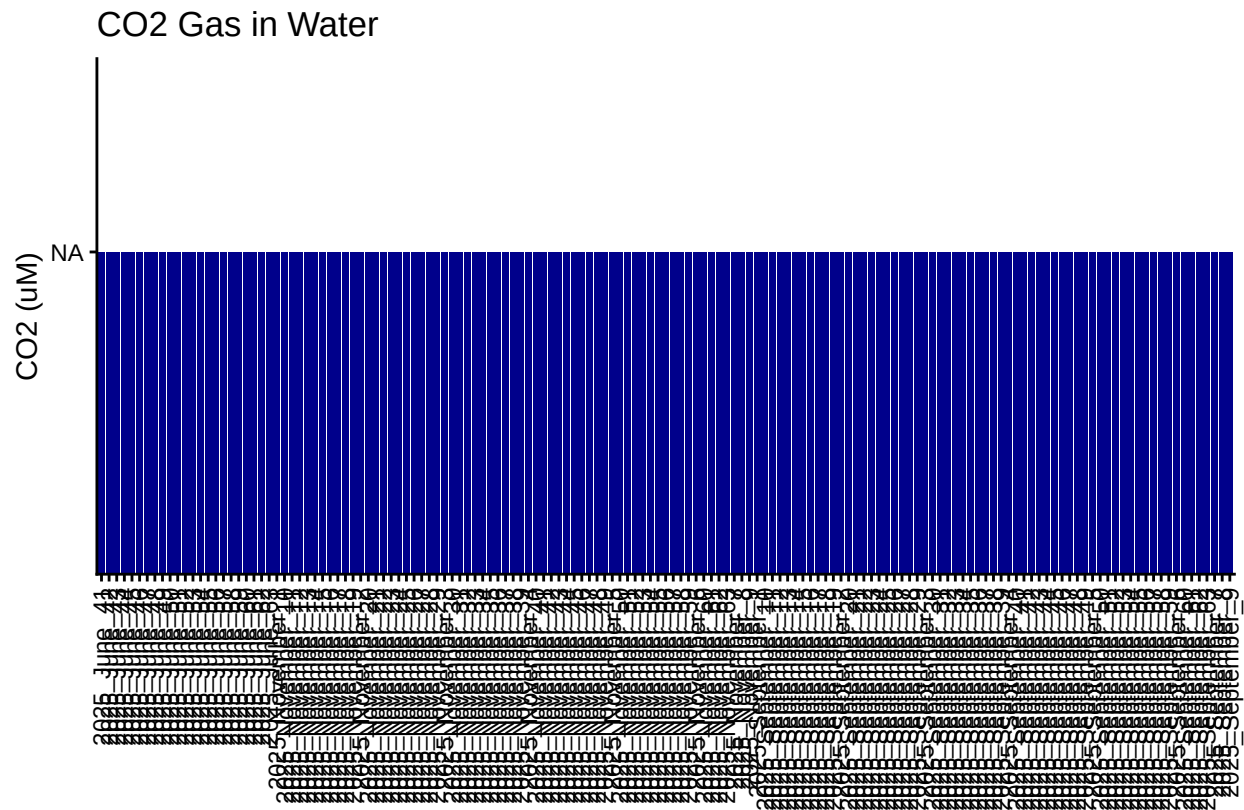
12 Check Range - N2O



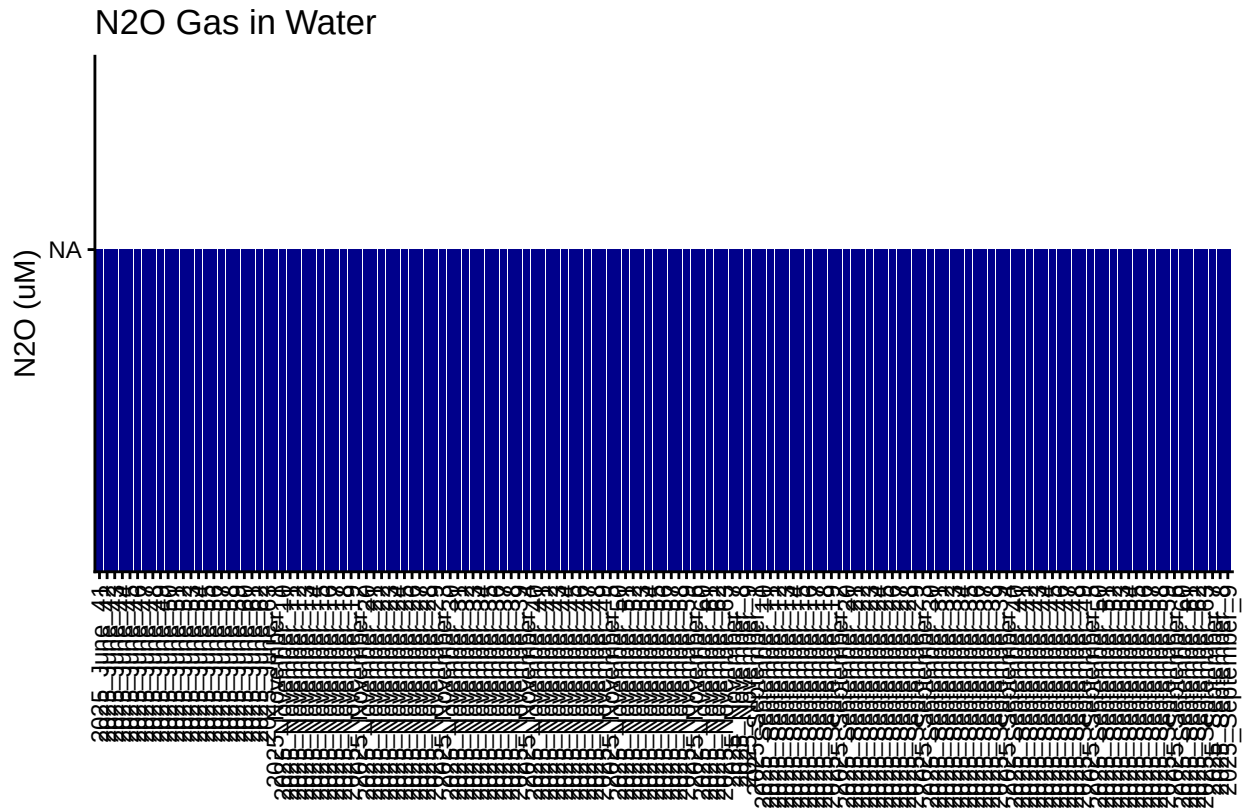
13 Calculate gas in water - CH₄



14 Calculate gas in water - CO2



15 Calculate gas in water - N2O



16 Merge samples with Metadata & check all samples are present

```
## Merge Samples with Metadata
```

```
## All sample IDs are present in data
```

16.1 Organize Data

```
##      Sample_ID  Type  Run_Date CH4_Area Conc_CH4      CH4_RTime_Flag CO2_Area
## 1 2025_June_41 Sample 2026-09-02    424      NA Correct - CH4 RTime    75028
## 2 2025_June_42 Sample 2026-09-02    713      NA Correct - CH4 RTime    45562
## 3 2025_June_43 Sample 2026-09-02    510      NA Correct - CH4 RTime    50626
## 4 2025_June_44 Sample 2026-09-02    453      NA Correct - CH4 RTime    37138
## 5 2025_June_45 Sample 2026-09-02    767      NA Correct - CH4 RTime    28019
## 6 2025_June_46 Sample 2026-09-02    453      NA Correct - CH4 RTime    48410
##      Conc_CO2      CO2_RTime_Flag N2O_Area Conc_N2O      N2O_RTime_Flag
## 1      NA Correct - CO2 RTime    301      NA Correct - N2O RTime
## 2      NA Correct - CO2 RTime    433      NA Correct - N2O RTime
## 3      NA Correct - CO2 RTime    326      NA Correct - N2O RTime
## 4      NA Correct - CO2 RTime    513      NA Correct - N2O RTime
## 5      NA Correct - CO2 RTime    330      NA Correct - N2O RTime
```

##	6	NA Correct - CO2 RTime	419	NA Correct - N2O RTime
##		FID_Peak_Flag	ECD_Peak_Flag	CH4_Conc_ppm CH4_Flag
##	1	Correct	Correct	0.6247897 BDL
##	2	Incorrect - Too Many Peaks	Correct	1.6886552 Within Range
##	3	Correct	Correct	0.9413725 Within Range
##	4	Correct	Correct	0.7315444 BDL
##	5	Correct	Correct	1.8874398 Within Range
##	6	Correct	Correct	0.7315444 BDL
##		CO2_Conc_ppm	CO2_Flag	N2O_Conc_ppm N2O_Flag CH4_Conc_uM CO2_Conc_uM
##	1	51110.56	ADL	-0.04153180 BDL NA NA
##	2	30826.50	Within Range	0.05086752 BDL NA NA
##	3	34312.50	Within Range	-0.02403193 BDL NA NA
##	4	25027.52	Within Range	0.10686710 BDL NA NA
##	5	18750.10	Within Range	-0.02123195 BDL NA NA
##	6	32787.04	Within Range	0.04106759 BDL NA NA
##		N2O_Conc_uM		
##	1	NA		
##	2	NA		
##	3	NA		
##	4	NA		
##	5	NA		
##	6	NA		
##		Sample1	Sample_Year	Sample_Month Sample_Day Sample_Time Time_Zone G_W
##	1	SGW	2025	September 8 1012 EDT G
##	2	SGW	2025	September 8 1013 EDT G
##	3	SGW	2025	September 8 1014 EDT G
##	4	SGW	2025	September 8 1015 EDT G
##	5	SGW	2025	September 8 1020 EDT G
##	6	SGW	2025	September 8 1021 EDT G
##		Sample_Plot	Sample_Grid	Sample_Depth Field_Notes Lab_Notes CH4_Conc_ppm
##	1	Control	C3	10 NA NA 1.1180699
##	2	Control	C3	20 NA NA 0.4996290
##	3	Control	C3	35 NA NA 0.4260051
##	4	Control	C3	50 NA NA 0.3855120
##	5	Control	C6	10 NA NA 1.3499852
##	6	Control	C6	20 NA NA 1.1401571
##		CH4_Conc_uM	CH4_Conc_Flag	CO2_Conc_ppm CO2_Conc_uM CO2_Conc_Flag N2O_Conc_ppm
##	1	NA	Within Range	2576.413 NA Within Range 0.13906686
##	2	NA	BDL	7527.304 NA Within Range 0.16216669
##	3	NA	BDL	8940.565 NA Within Range 0.13556689
##	4	NA	BDL	15253.777 NA Within Range 0.15166677
##	5	NA	Within Range	3563.562 NA Within Range 0.07746732
##	6	NA	Within Range	10532.120 NA Within Range 0.17056663
##		N2O_Conc_uM	N2O_Conc_Flag	QAQC_Flags
##	1	NA	BDL	N2O_R2 is Bad; N2O Check Standards out Of Range
##	2	NA	BDL	N2O_R2 is Bad; N2O Check Standards out Of Range
##	3	NA	BDL	N2O_R2 is Bad; N2O Check Standards out Of Range
##	4	NA	BDL	N2O_R2 is Bad; N2O Check Standards out Of Range
##	5	NA	BDL	N2O_R2 is Bad; N2O Check Standards out Of Range
##	6	NA	BDL	N2O_R2 is Bad; N2O Check Standards out Of Range
##		Analysis_rundate		Run_notes
##	1	2026-09-02	Run 9/02 - Blank1,2,4,5,11 -> ChkStd -- Corrected	
##	2	2026-09-02	Run 9/02 - Blank1,2,4,5,11 -> ChkStd -- Corrected	

```
## 3      2026-09-02 Run 9/02 - Blank1,2,4,5,11 -> ChkStd -- Corrected
## 4      2026-09-02 Run 9/02 - Blank1,2,4,5,11 -> ChkStd -- Corrected
## 5      2026-09-02 Run 9/02 - Blank1,2,4,5,11 -> ChkStd -- Corrected
## 6      2026-09-02 Run 9/02 - Blank1,2,4,5,11 -> ChkStd -- Corrected
```

16.2 Visualize Data CH4

```
## Visualize Data CH4
```

TEMPEST: Soil Gas Well CH4



16.3 Visualize Data CO2

```
## Visualize Data CO2
```

TEMPEST: Soil Gas Well CO2



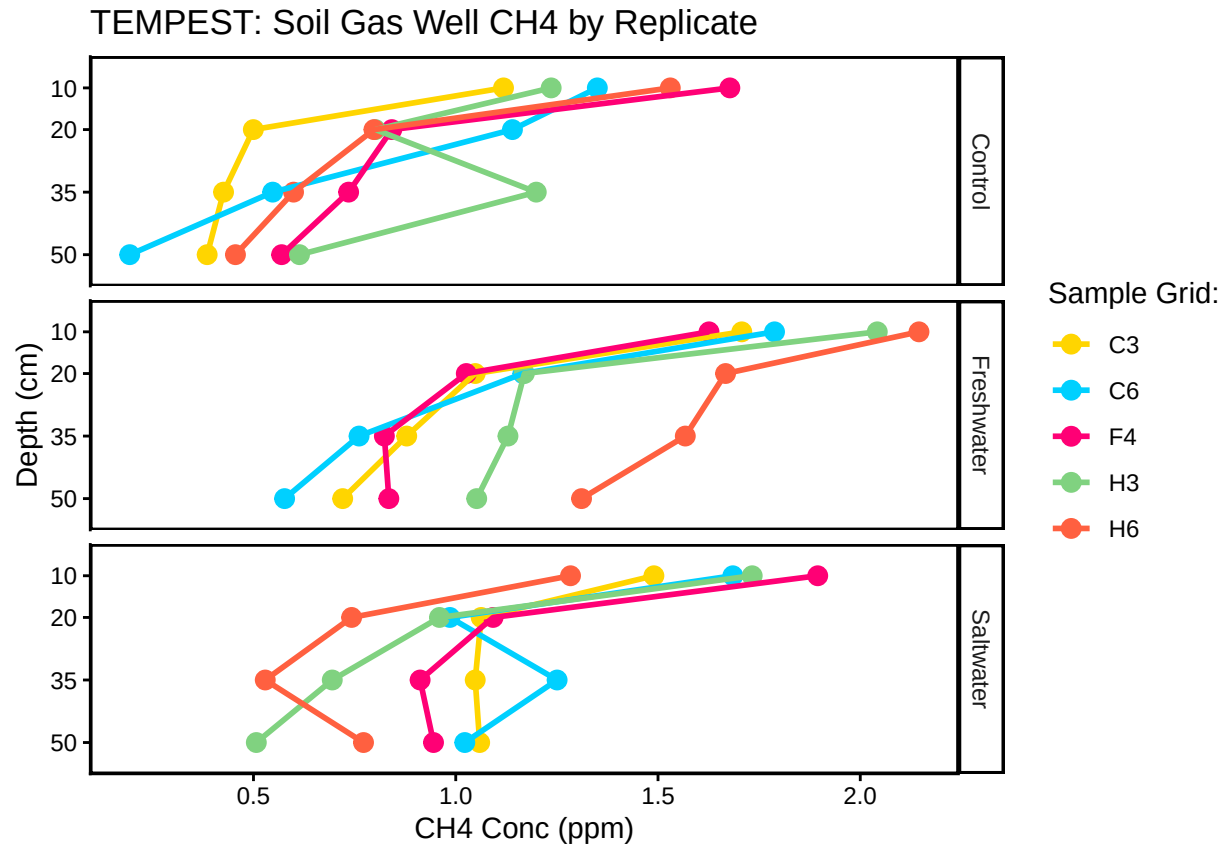
16.4 Visualize Data N2O

```
## Visualize Data N2O
```

TEMPEST: Soil Gas Well N2O

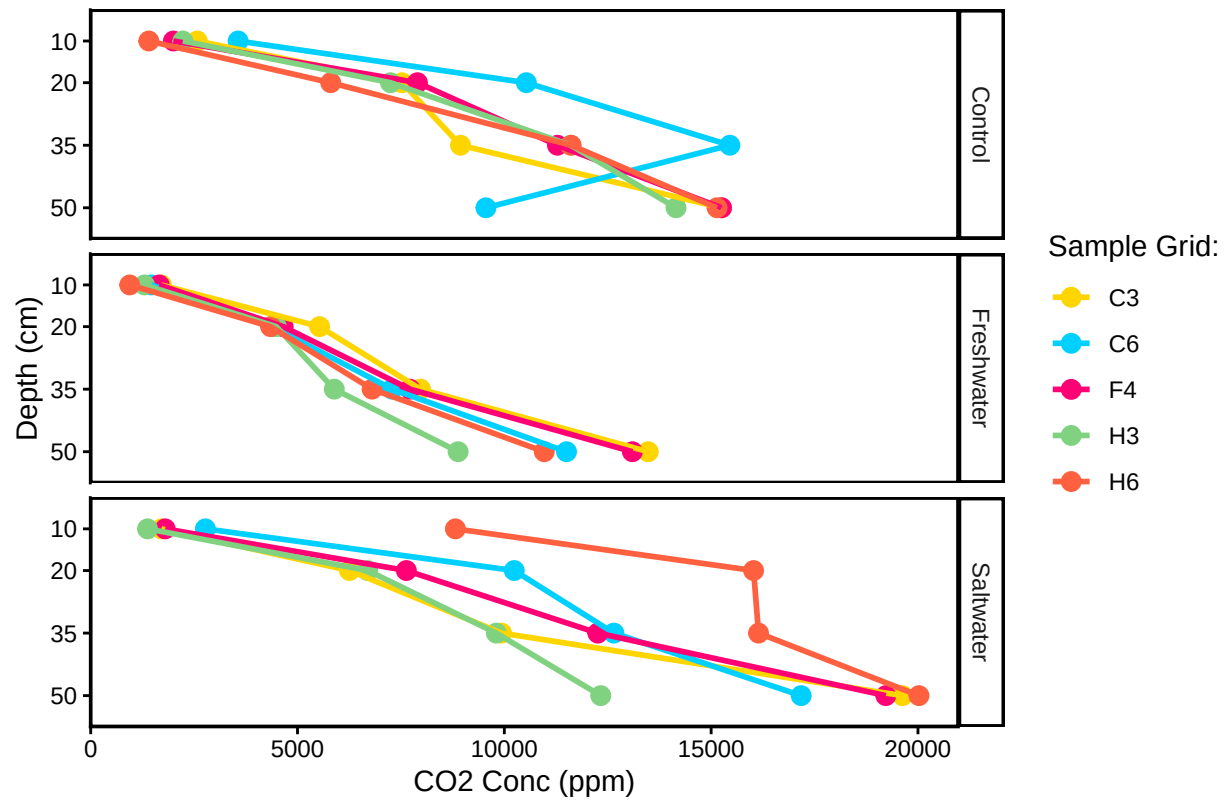


#Visualize Data CH4 by Replicate



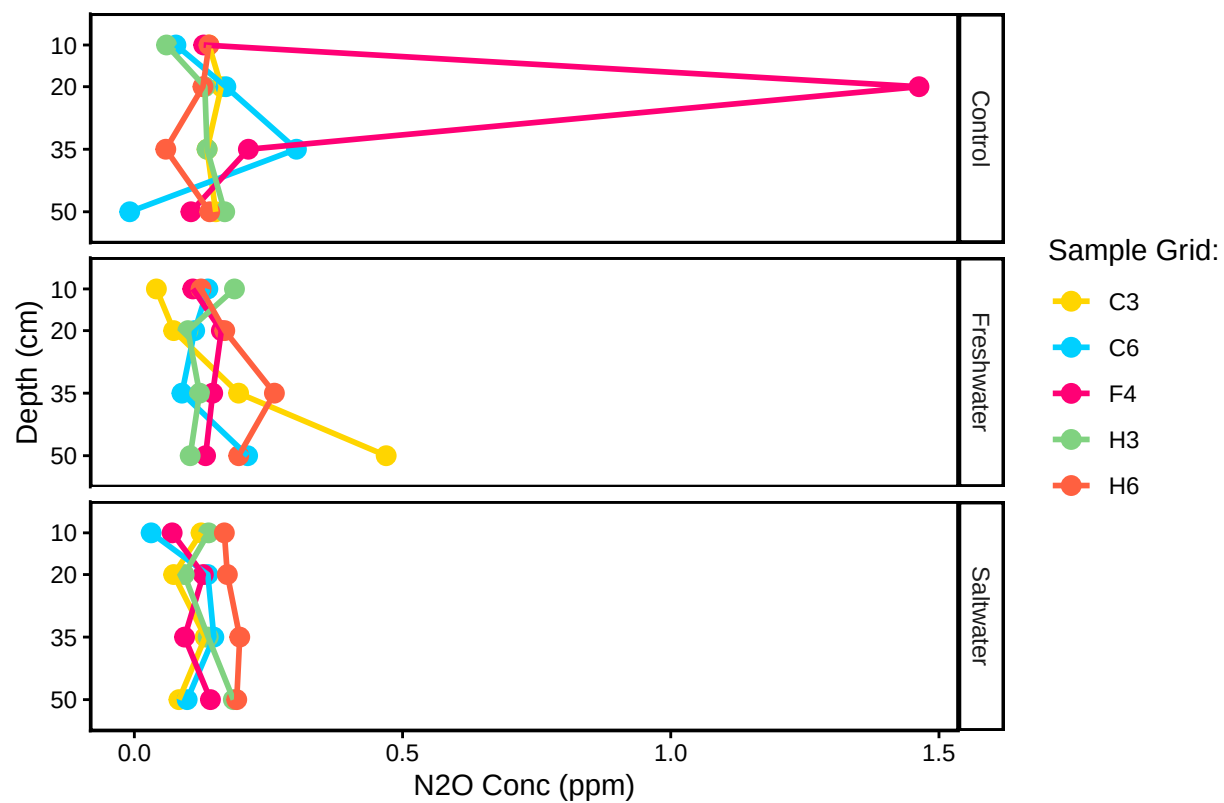
#Visualize Data CO2 by Replicate

TEMPEST: Soil Gas Well CO2 by Replicate



#Visualize Data N2O by Replicate

TEMPEST: Soil Gas Well N2O by Replicate



##Export Data

#end